A gentle introduction to the file system and the user interface

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Why to bother about Command Line Interfaces?

- `Cleaner approach`: Command line skills help with building repeatable data processes
- `Repeatability`: They facilitate repeatable data manipulation
- `Safer`: Performing repetitive tasks in a CLI makes your work less error prone
- `Precision`: Many bioinformatics tools can be used only by means of a CLI or their CLI version give you access to many more parameters.
- `HPC/Cloud`: Computationally intense work can be only performed on remote or cloud resources, and the preferred access to them is the CLI

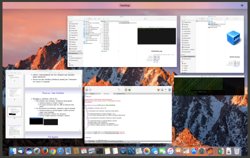
CLI and File System

Most of the time spent on the CLI is used to run programs and manage files.

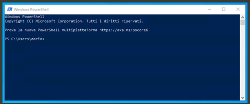
In this mini course you will learn how to use the CLI and how to use it to move in the file system

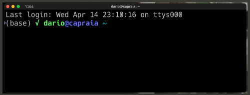
User Interface: operating system access

Two paradigms: GUI vs CLI



- GUI (Graphic User Interface).
  - the interaction is based on windows, buttons, ...
  - it is the most recent (created in the '80s in the Xerox PARC labs)



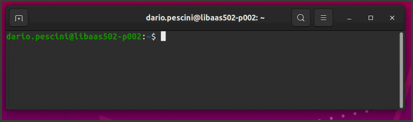


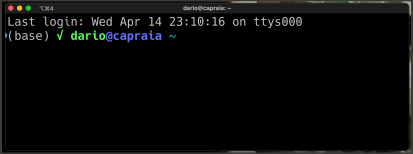
- CLI (Command Line Interface) or Shell.
  - the interaction is based on commands, typed on a keyboard
  - the output: on a screen, in a file, on a printer, ...
  - Useful to execute complex and repetitive actions
  - Philosophy: each specific command do a specific job, combine them into scripts to solve complex problems

Modern OSs: a shell inside a window

Different OSs, different Shells:

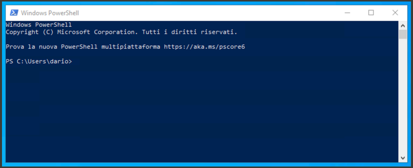
- Linux `terminal`: (Bash, zsh, ...)



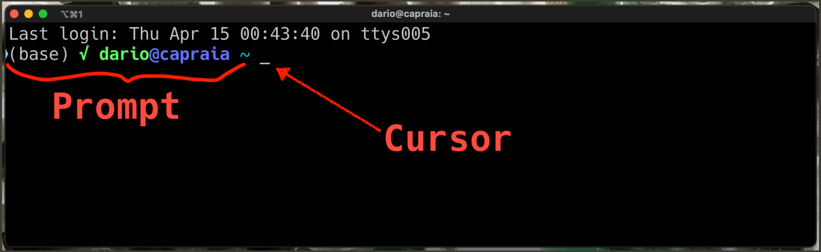


- OS X `terminal/iterm`: (zsh, Bash, ...)

- Windows `PowerShell/cmd.exe`

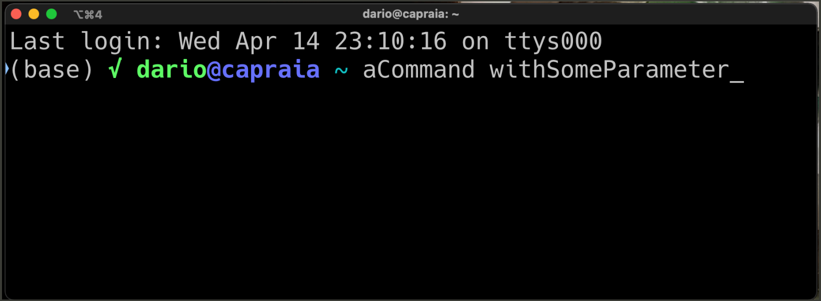


How to interact with the shell



- the shell `prompt` waits you to insert a command at the `cursor`

How to interact with the shell

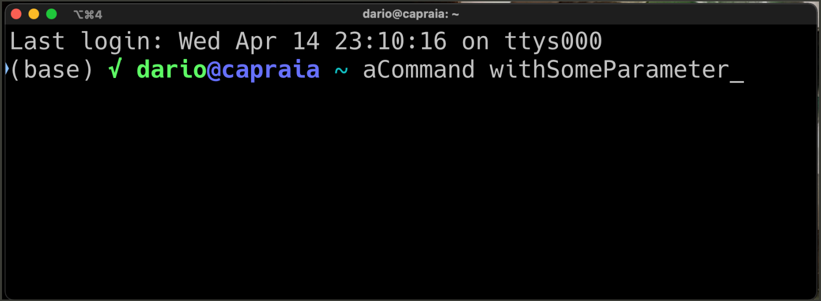


- a command has this pattern:

  command [parameter_1] [parameter_2]... [parameter_N]

- where everything between [and] is optional

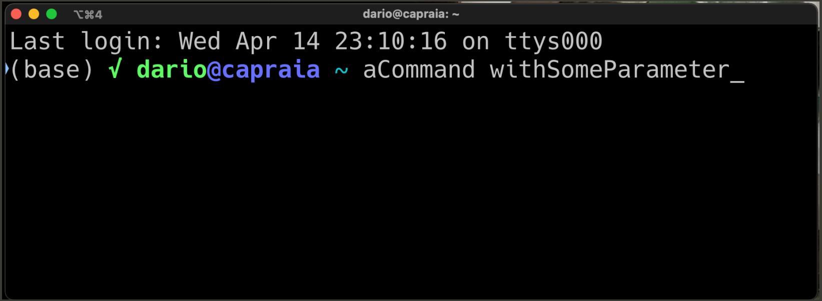
How to interact with the shell



- a command has this pattern:

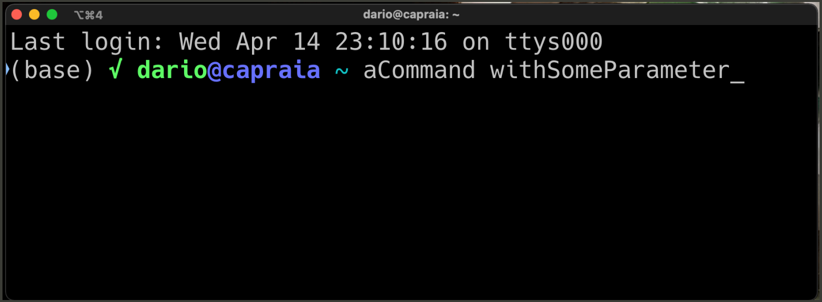
  command [parameter_1] [parameter_2]... [parameter_N]
- where everything between [and] is optional
- manual: `man command` open the documentation for the command

How to interact with the shell



- tab completion: start to digit a command and then press ⇥ (tab)

How to interact with the shell



- tab completion: start to digit a command and then press ⇥ (tab)
- history: it is possibile to navigate the already used commands by means of ↑ and ↓

The File System

The `File System` is the Operating System (OS) component dedicated to the `permanent storage` of the information.

- It controls `how` the information is stored and accessed in storage disks
- It exposes to the user simple interfaces to `manage` (write, delete, copy, ...) the stored information

File system

From a user point of view, it is built upon two building blocks:

- `file` : uniform logic representation of correlated information
  - atomic entity
  - collection of correlated information
- `directory` : collection of information used to organize and access files meta-data
  - collection of files
  - access point to files

File Attributes - meta data

- `name`: A string used to identify a specific file within the file system
- `location`: the position of the file on the storage unit
- `kind`: It is used by some OSs to asses the kind of the file content
- `dimension`
- `date-time`: information relatives to time of the creation or last modification
- `ownership`: owner, group. Used to manage authorizations and accounting
- `protection`: file access policies

File, Directory and Meta dati

When you list the content of a folder (i.e. directory) the shell exposes you files and folders meta data:

```
(base) √ dario@capraia dario ls -lh
total 872
drwxr-xr-x  2 dario  staff    64B Apr 15 22:25 A01
drwxr-xr-x  2 dario  staff    64B Apr 15 22:25 A02
-rw-rw-r--  1 dario  staff   416K Apr 15 22:25 CFCodiceComuniItalia.csv
-rw-rw-r--  1 dario  staff   233B Apr 15 22:25 CFTabellaDispari.tsv
-rw-rw-r--  1 dario  staff   228B Apr 15 22:25 CFTabellaPari.tsv
-rw-rw-r--  1 dario  staff   119B Apr 15 22:25 CFTabellaResto.tsv
-rw-------  1 dario  staff   4.0K Apr 15 22:25 CFcalcolo.py
```

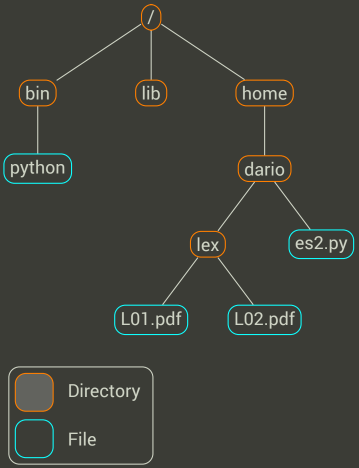
dir	Owner Perm.	Group Perm.	Others Perm.	Owner Name	Group Name	Size	Last Modified	Name
d	rwx	r-x	r-x	dario	staff	64B	Apr 15 22:25	A01

Permissions:

- d directory
- r read
- w write
- x execute/traverse

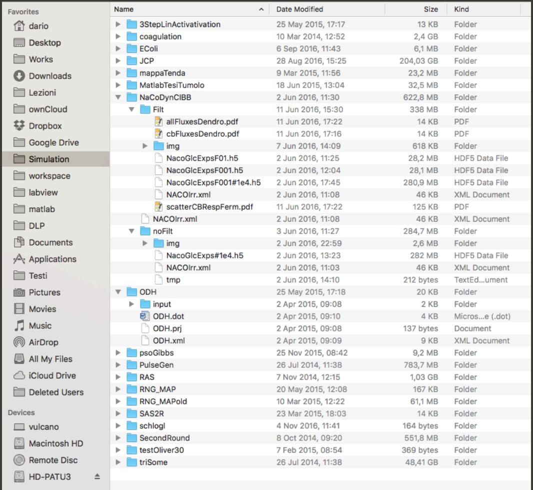
File System

In the file system, files and folders are structured as a tree:



File System

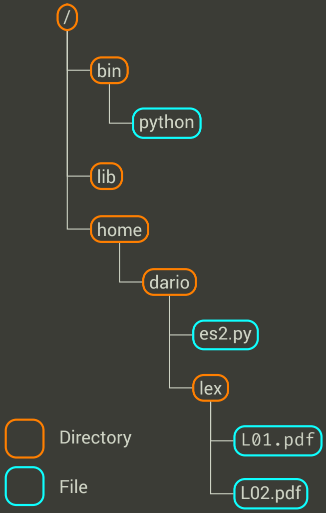
Also from the user perspective the ”file system” is structured as a tree:



Name	Date Modified	Size	Kind
3StepLinActivativation	25 May 2015, 17:17	13 KB	Folder
coagulation	10 Mar 2014, 12:52	2,4 GB	Folder
EColi	6 Sep 2016, 11:43	6,1 MB	Folder
JCP	28 Aug 2016, 15:25	204,03 GB	Folder
mappaTenda	9 Mar 2015, 11:56	23,2 MB	Folder
MatlabTesiTumolo	18 Jun 2015, 13:04	32,5 MB	Folder
NaCoDynCIBB	2 Jun 2016, 11:30	622,8 MB	Folder
Filt	11 Jun 2016, 15:30	338 MB	Folder
allFluxesDendro.pdf	11 Jun 2016, 17:22	14 KB	PDF
cbFluxesDendro.pdf	11 Jun 2016, 17:16	14 KB	PDF
img	7 Jun 2016, 14:09	618 KB	Folder
NacoGlcExpsF01.h5	2 Jun 2016, 11:25	28,2 MB	HDF5 Data File
NacoGlcExpsF001.h5	2 Jun 2016, 12:04	28,1 MB	HDF5 Data File
NacoGlcExpsF001#1e4.h5	2 Jun 2016, 17:45	280,9 MB	HDF5 Data File
NACOlrr.xml	2 Jun 2016, 11:08	46 KB	XML Document
scatterCBRespFerm.pdf	11 Jun 2016, 17:22	125 KB	PDF
NACOlrr.xml	2 Jun 2016, 11:08	46 KB	XML Document
noFilt	3 Jun 2016, 11:27	284,7 MB	Folder
img	2 Jun 2016, 22:59	2,6 MB	Folder
NacoGlcExps#1e4.h5	2 Jun 2016, 13:23	282 MB	HDF5 Data File
NACOlrr.xml	2 Jun 2016, 11:03	46 KB	XML Document
tmp	2 Jun 2016, 14:10	212 bytes	TextEd...ument
ODH	25 May 2015, 17:18	20 KB	Folder
input	2 Apr 2015, 09:08	2 KB	Folder
ODH.dot	2 Apr 2015, 09:10	4 KB	Micros...e (.dot)
ODH.prj	2 Apr 2015, 09:08	137 bytes	Document
ODH.xml	2 Apr 2015, 09:09	9 KB	XML Document
psoGibbs	25 Nov 2015, 08:42	9,2 MB	Folder
PulseGen	26 Jul 2014, 11:38	783,7 MB	Folder
RAS	7 Nov 2014, 12:15	1,03 GB	Folder
RNG_MAP	20 May 2015, 12:08	167 KB	Folder
RNG_MAPold	10 Mar 2015, 12:22	61,1 MB	Folder
SAS2R	23 Mar 2015, 18:03	14 KB	Folder
schlogl	4 Nov 2016, 11:41	164 bytes	Folder
SecondRound	8 Oct 2014, 09:20	551,8 MB	Folder
testOliver30	7 Feb 2015, 08:54	369 bytes	Folder
triSome	26 Jul 2014, 11:38	48,41 GB	Folder

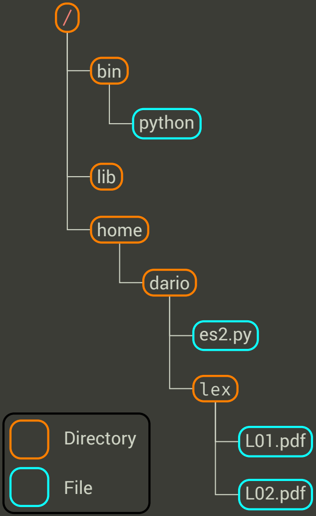
File system: jargon

Given the file `L01.pdf` in the file system on the right,



- L01.pdf is the file **name** (or base name)
- pdf is the file **extension**
- /home/dario/lex/ is the **directory** that contains the file
- /home/dario/lex/L01.pdf is the **absolut path** (*fully qualified name* or *full path*)
- **relative path**
  - given that /home/dario/ is the current **working directory**
  - lex/L01.pdf is the **relative path** of L01.pdf w.r.t. the working directory

File system: jargon



- It is possible to reason about folders name and path, in an analogous way to what we have done for files.
- Given the path /home/dario/lex/L01.pdf
  - / is the **root**, folder: the origin of every and each path in the file system
  - the other / are file and folders visual separators
  - the symbol / is called *slash*

File Systems in Unix and Windows

These two families of OSs use different folder separator / slash versus \ back slash, this can hamper your work.

OS	Root Directory	Directory Separator	Examples
Unix	/	/	/home/user/docs/Letter.txt
Win	[drive letter:]\	\	C:\User\MrX\Letter.txt \User\MrX\Letter.txt MrX\Letter.txt

- The [ ] wrapping the *drive letter:* means that you can omit the Drive letter indication

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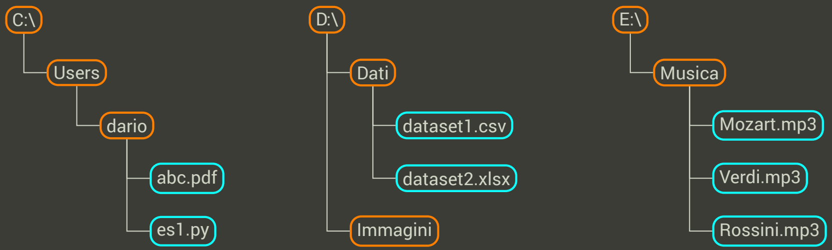
Win	[drive letter:]\	\	C:\User\MrX\Letter.txt \User\MrX\Letter.txt MrX\Letter.txt

- The [ ] wrapping the *drive letter:* means that you can omit the Drive letter indication
- If in windows you use the PowerShell you can ignore this difference.

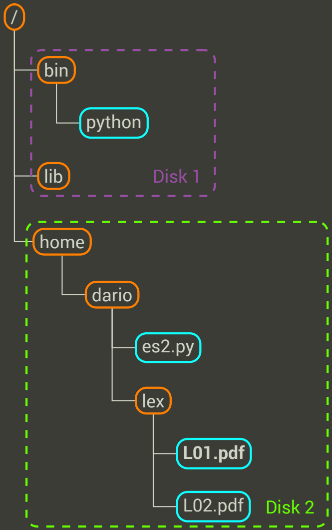
Windows: Drives and Forests



- The windows file system is a forest of trees
- Each drive is the root of a distinct tree.
- To change the tree you have to call it: `E:`

Unix: Drives on a Tree

- In the Unix file systems, apparently there is no trace of the different drives.
- The devices are directly *mounted* on the folders tree.
- This transparent approach make it easier to move folders to different drives.



How to navigate in your file system

- `pwd` - where am I ?
- `ls` - what is in here ?
- `cd` - take me there !

Paths recap

- / root
- . here
- ~ home
- .. level up
- - where I was before